MIL-P-48323C (AR)
19 November 1984
SUPERSEDING
MIL-P-48323B (AR)
22 May 1981

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, RA, M650 ROCKET MOTOR BODY ASSEMBLY

This specification is approved for use by US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This Military Specification covers the Rocket Motor Body Assembly and components for Projectile, 8 Inch, HE, Rocket Assisted, M650.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pre-treatment of Ferrous Surfaces for Original Coatings
PPP-B-636	- Boxes, Shipping, Fiberboard
TT-I-1795	- Ink, Marking, Stencil, Opaque (porous and non-porous surfaces)

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-packaging, Methods Of
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packaging
MIL-A-2550	- Ammunition, General Specification For
MIL-I-6868	- Inspection Process, Magnetic Particle
MIL-R-13562A	- Rotating Bands, Welded Overlay, Ferrous and Nonferrous, for Projectiles
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

FED-STD-141	- Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing
FED-STD-151	- Metals; Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-410	- Non Destructive Testing Personnel Qualification and Certification
MIL-STD-453	- Inspection Radiographic

2.1.2 Other government documents, drawings and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGE DRAWINGS

8837835	- Pallet for Projectiles and Projectile Metal Parts
8861520	- Marking Diagram for Pallets for Projectiles 75MM t0 120MM
9288025	- Rocket Motor Body Assembly

INSPECTION EQUIPMENT DRAWINGS

9280261	- Magnetic Particle Standard for Motor Body (Envelope Drawing)
9280353	- Envelope Drawing C-Scan M650 Motor Body

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E8 - Tension Testing of Metallic Materials
E10 - Brinnell Hardness of Metallic Materials, Test Method for
E103 - Rapid Indentation Hardness Testing of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103).

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, NY 10018).

PANGBORN ROTO BLAST STEEL SHOT AND GRIT CATALOG

(Copies may be obtained from Pangborn Division of the Carborundum Co., Hagerstown, Md. 21740).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with drawing 9288025 and associated drawings and with all requirements specified in applicable specifications and standards.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9288025 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Chemical requirement. The iron content of the copper welded overlay shall be 0.25 minimum to 3.000 maximum percent by weight.

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3.4 Metal defects.

3.4.1 Motor body. There shall be no cracks, seams, laminations and laps on the motor body surface, except that indications no greater than .100 inch are permitted in the undercut forward of the copper welded overlay and 1/8 inch in the undercut aft (including angled surface) of the copper welded overlay. No crack in either undercut shall have a sharp curve.

3.4.2 Copper welded overlay.

3.4.2.1 Cracks. There shall be no cracks on the surface of the finish machined welded overlay.

3.4.2.2 Seams. There shall be no seams, extending across more than three crests, on the surface of the finish machined welded overlay.

3.4.2.3 Surface voids. Voids at the surface of the finish machined welded overlay shall comply with applicable requirements of MIL-R-13562A.

3.4.2.4 Iron deposits. There shall be no iron deposits in excess of 1/8 inch in diameter.

3.4.3 Copper welded overlay seat. There shall be no cracks, seams, laminations and laps on the copper welded overlay seat (see 6.9). Except, indications due to intergranular penetration shall comply with the visual standards submitted in accordance with 3.4.5.

3.4.4 Copper welded overlay seat (subsurface). There shall be no cracks, seams, laminations and laps at the 7.810 diameter below the copper welded overlay seat, except for the starting weld zone (see 6.8). There shall be no indications due to intergranular penetration at the 7.770 inch diameter.

3.4.5 Visual standards. Visual standards for evaluation of indications due to intergranular penetration in accordance with 3.4.3 shall be established at the time of first article sample submission.

3.5 Base metal cratering.

3.5.1 Copper welded overlay seat. The frequency and distribution of cratering shall comply with the visual standards submitted in accordance with 3.5.3.

3.5.2 Copper welded overlay seat (subsurface). There shall be no cratering, except for the starting weld zone, at the 7.840 inch diameter below the copper welded overlay seat. There shall be no cratering at the 7.770 inch diameter.

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3.5.3 Visual standards. Visual standards for evaluation of cratering in accordance with 3.5.1 shall be established at the time of first article sample submission.

3.6 Ultrasonic inspection. Ultrasonic c-scan inspection shall be utilized to inspect for quality of bond of the rotating band after heat treat.

3.6.1 Linear requirement. Linear measurements shall be made with a scale having 1/16 inch graduations. The maximum linear dimensions of any unbonded area shall be one inch maximum. Thin, pointed spikes shall be disregarded. Additional definition of this requirement is made in Figure 1.

3.6.1.1 Vertical unbond areas. A vertical unbond is an area whose length runs across the width of the rotating band. One vertical unbond across the width of the band is acceptable if the average width is 1/2 inch maximum and the maximum width is 3/4 inch (See Figure 2). One vertical unbond which does not reach one or both edges of the band shall be acceptable if the maximum width is 3/4 inch (See Figure 2). Only one unbond area in any band may be accepted under this definition.

3.6.1.2 Edge unbond areas. An unbond extending along the entire edge of the rotating band is acceptable if its maximum width is 3/16 inch, disregarding minor bulges (See Figure 3). Edge unbonds of 2 inches maximum length and 1/4 inch maximum width are acceptable if separated by 1/4 inch minimum (See Figure 3).

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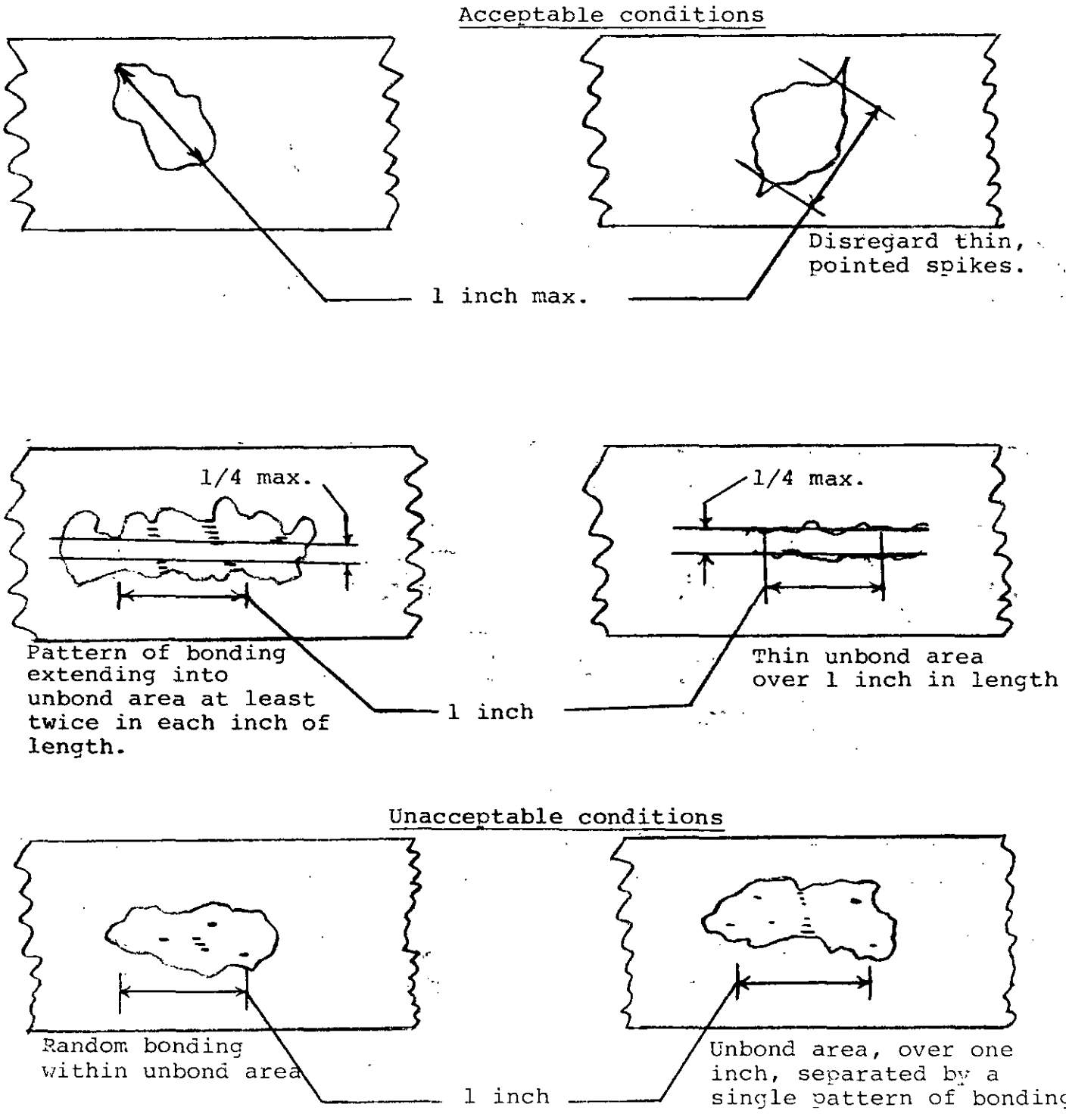


FIGURE 1. Maximum linear dimension of unbond

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Acceptable conditions

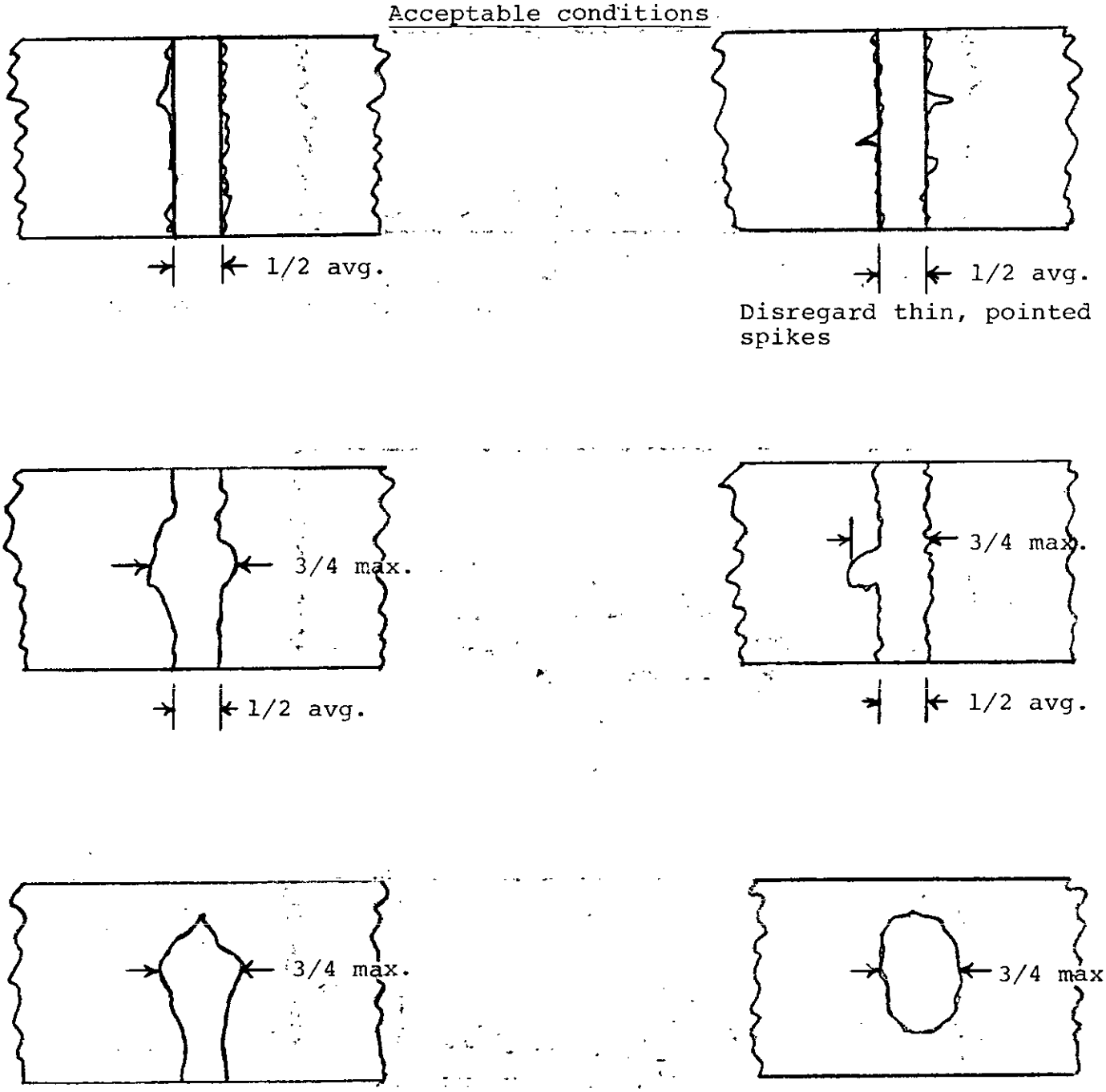
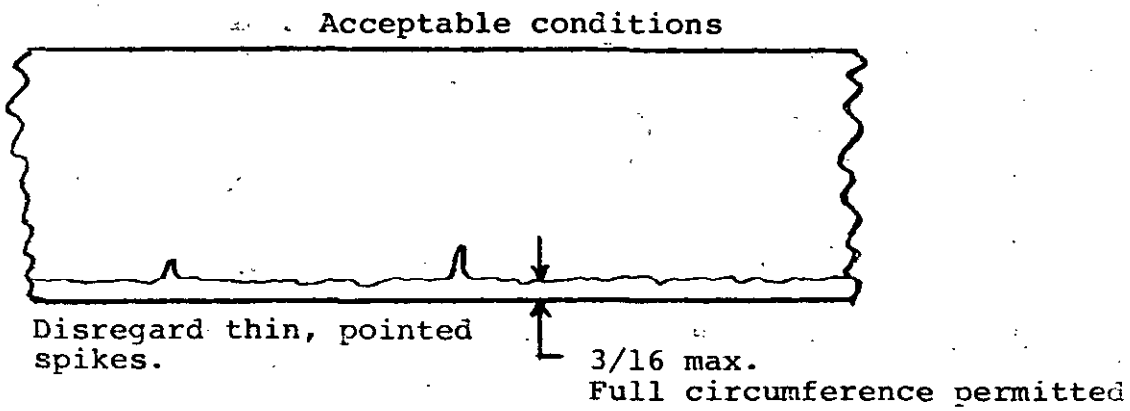
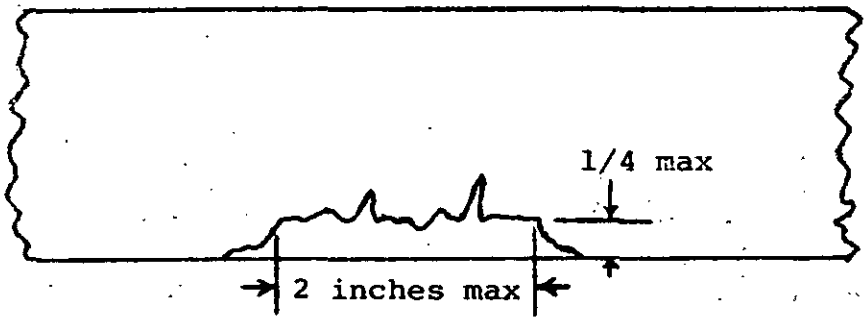


FIGURE 2. Veritcal unbond areas

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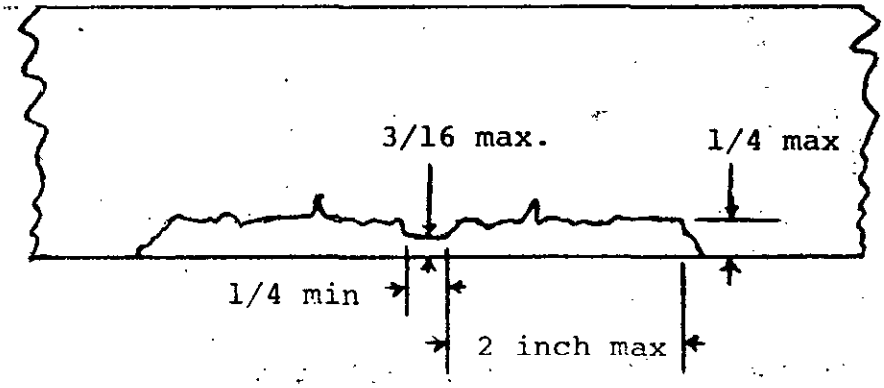
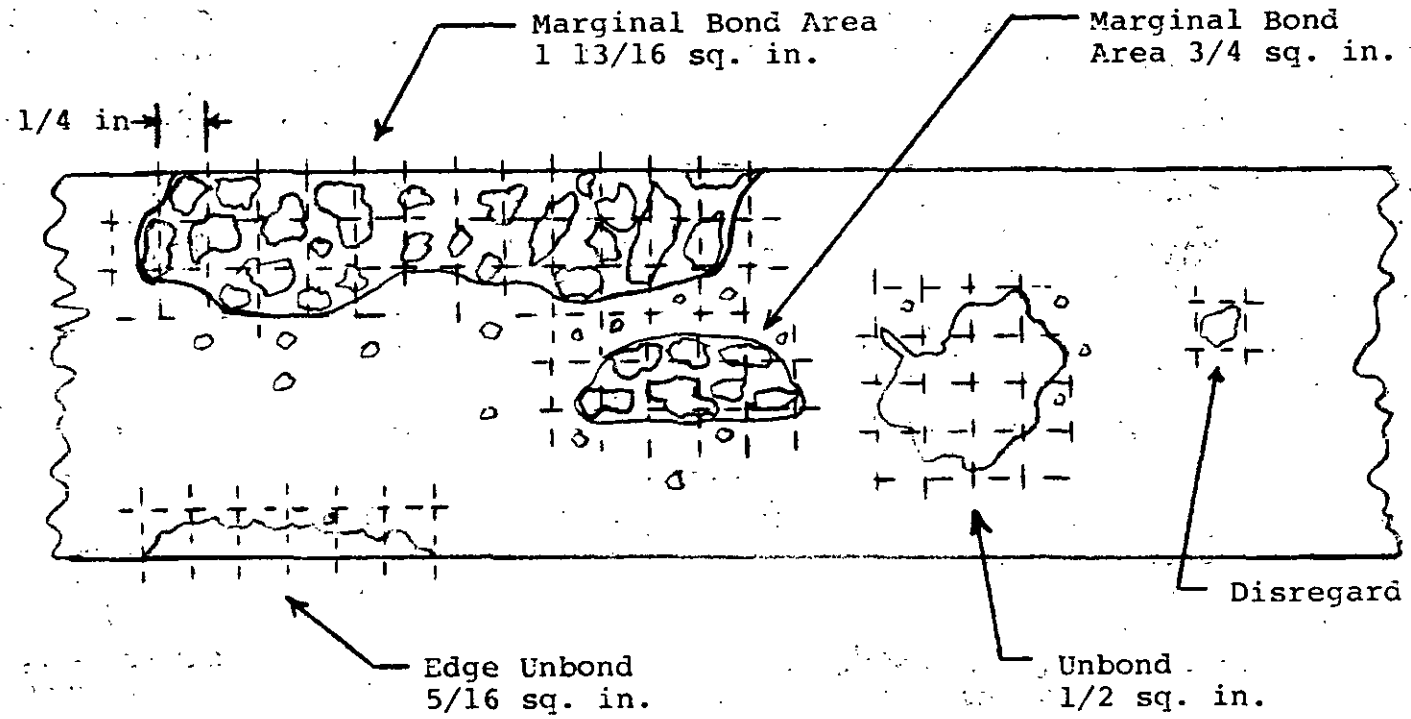


FIGURE 3. Edge unbond areas

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3.6.2 Area requirements. The maximum total area of unbonds and marginal bonds shall be 12 square inches. A marginal bond area is any pattern of small unbonds. A clear plastic template with 1/4 inch square grids shall be used to visually sum unbonded and marginally bonded areas (See Figure 4). Unbonds of less than 1/16 square inch, that are not part of a pattern, shall be disregarded.

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Notes:

1. Total area of illustrated unbonds and marginal bonds is 3 3/8 square inches (sq. in.).
2. The border of a marginal bond area along the edge of the rotating band shall be the edge.
3. All grid squares within the border of a marginally bonded area shall be counted.
4. Grid squares along the border of an unbonded or marginally bonded area shall be counted if the border includes 1/2 or more of the square.

FIGURE 4. Summing unbonds and marginal bonds

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3.6.3 Automated acceptance requirements. For automated ultrasonic c-scan interpretation, no unbond area shall exceed 1/4 x 3/4 inches either perpendicular or parallel to the edges of the band. In addition, the total of unbonded areas shall not exceed six square inches.

3.6.4 Motor body I.D. prior to ultrasonic inspection. The inside diameter prior to ultrasonic inspection shall be of a contractor specified diameter with tolerance of .015 inch total with a surface finish of 125 RMS or better in the area of the rotating band. This requirement is to assure that the contractor will inspect to the same I.D. each time. The I.D. in this case may be prior to final machining. Also, the corner radius at the bulkhead shall be such as to not interface with the c-scan inspection in the area of the band.

3.7 Proving ground test. The motor body bonding assembly shall not collapse, separate or break up in the gun bore or in flight. Testing shall be as specified in 4.5.6.

3.8 Nozzle insert.

3.8.1 Compression test. The nozzle insert shall withstand a compression load of 5000 pounds, minimum without cracking or breaking.

3.8.2 X-ray. (See Figure 5).

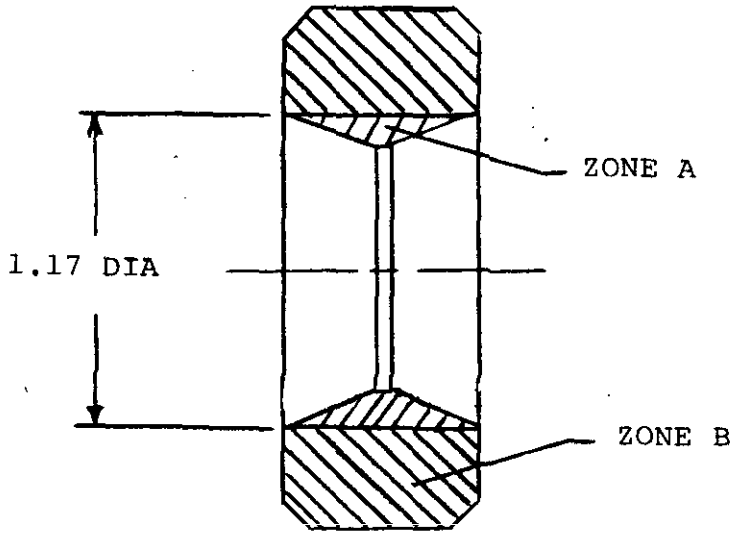
Zone A - There shall be no voids greater than 0.030 inch in diameter.

Zone B - There shall be no voids greater than 0.060 inch in diameter. There shall be no voids greater than 0.030 diameter which are closer than 0.060 inch apart.

3.9 Grain support. The grain support shall have no voids.

3.10 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI - B46.1 and Pangborn Shot and Grit Catalog shall be used as a basis for surface roughness determination.

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INSERT, NOZZLE (X-RAY)

FIGURE 5 Nozzle x-ray

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3.11 Protection of copper welded overlay. The copper welded overlay shall be protected from damage that may occur during manufacture and handling.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.13 Knurl. The minimum effective depth of the radial grooves in the knurl shall be .012 inches between the first and sixth peaks (See Figure 6).

3.14 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use.

3.15 Workmanship. The requirements for workmanship are as shown on the applicable drawing in the applicable specification and the following.

3.15.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.15.2 Painting. Painting shall comply with the requirements of the applicable drawings. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with drawing 9287971.

3.15.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of MIL-A-2550.

3.15.4 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.15.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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Reference Dwg. 9287971, "Informational View"

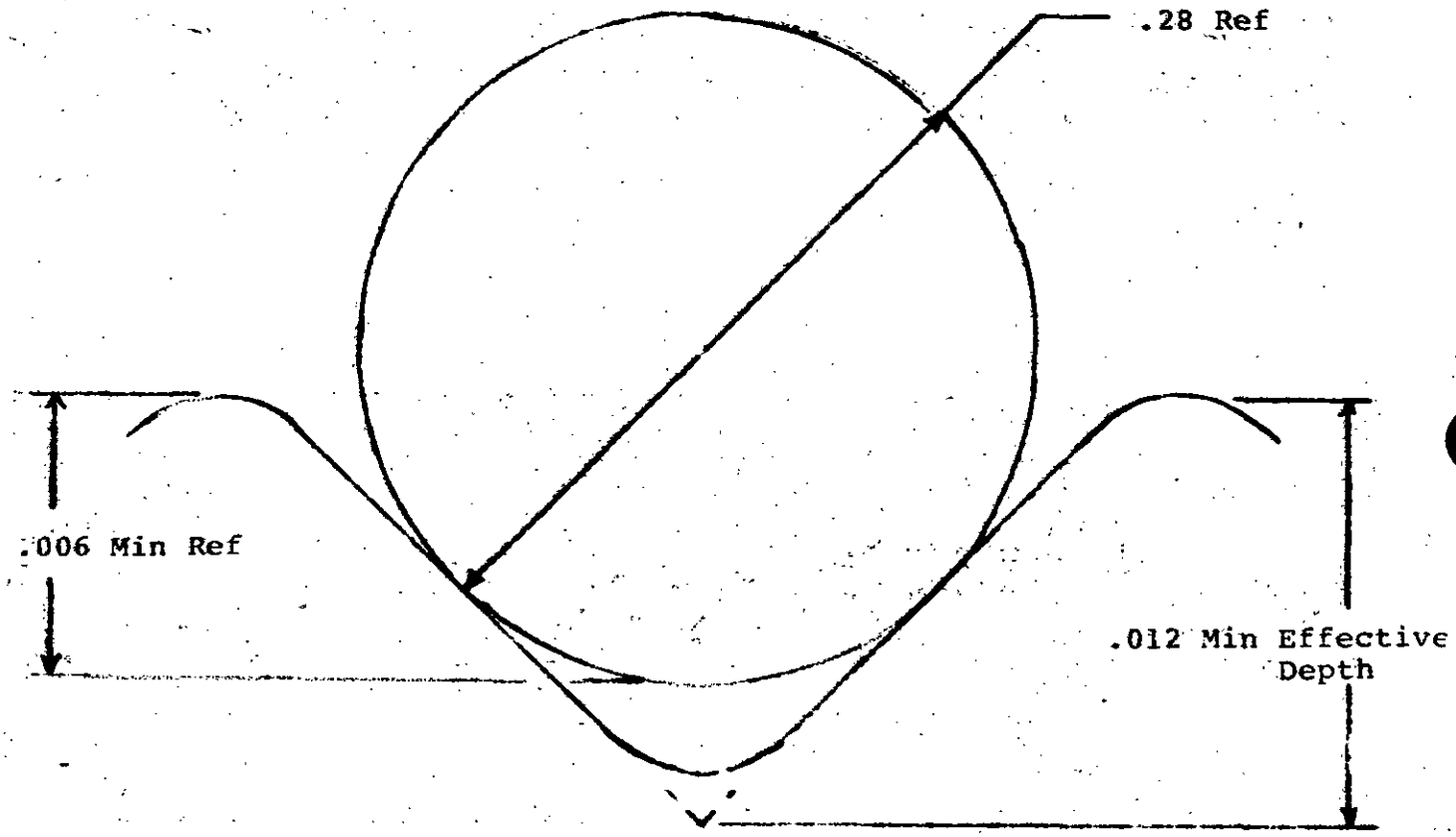


FIGURE 6 Knurl inspection

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3. First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Motor Body	9287971	20
Motor Body (Unknurled and unpainted)	9287971	10
Nozzle Insert	9287963	20
Unbanded Motor Body Slug	9287971	5
Grain Support	9287964	10 Consecutively produced from each cavity of each mold
Cushion Aft	9287986	10 Consecutively produced from each die
Motor Body Assy.	9288025	10
Motor Body Assy. (For Ballistic Testing, See 4.3.2.1)	9288025	20

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic tests.

4.3.2.1.1 Submission. Twenty complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.2).

4.3.2.1.2 Tests to be performed and acceptance criteria. Proving ground tests will be performed on twenty (20) assemblies in accordance with 4.4.3.8, utilizing the procedures of 4.5.6.to determine compliance with 3.7.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Motor Body Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Motor Body (Dwg. 9287971)				
	Examination for Defects	10	-	3.2	4.4.2.3
	Salt Spray Test	1	-	3.2	4.5.9
	Motor Body (Prior to Painting) (Dwg. 9187971)				
	Examination for Defects	10	-	3.2	4.4.2.2
	Magnetic Particle Test	10	-	3.4	4.5.3
	Ultrasonic Inspection	10	-	3.6	4.5.4
	Mechanical Properties of Motor Body	A	-	3.2	4.5.1
	Chemical Analysis	A	-	3.3	4.5.2
	Nozzle Insert (Dwg. 9287963)				
	Examination for Defects	20	-	3.2	4.4.2.4
	Nozzle Insert Tests	B	-	3.8	4.5.8
	Motor Body, Prior to Banding (Dwg. 9287971)				
	Examination for Defects	5	-	3.2	4.4.2.1
	Grain Support (Dwg. 9287964)				
	Examination for Defects	10	-	3.2	4.4.2.5

NOTES: A. Only one Mechanical Properties and Rotating Band Test sample shall be pulled from the different motor body configurations if they are all from the same lot.
B. Only one Nozzle Insert Test sample shall be pulled if all nozzle inserts are from one lot.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cushion, Aft (Dwg. 9287986)				
	Examination for Defects	10/ die	-	3.2	4.4.2.6
	Motor Body (Dwg. 9288025)				
	Examination for Defects	10	-	3.2	4.4.2.7, 4.4.2.8
	Proving Ground Test	20	-	3.7	4.5.7

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition to these provisions the lot formation provisions of 4.4.1.1 apply.

4.4.1.1 Heat treatment inspection lot. Heat treatment inspection lots shall not exceed one mill heat of steel. Whenever a break in heat treating of eight (8) hours or more occurs or a furnace is shut down for any length of time the heat treatment inspection lot will be terminated and a new lot shall start. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into larger heat treatment inspection lots after qualifying with individual batches provided qualifying procedures are submitted to and approved by the Government and provided all the other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Motor Body, Prior to Banding				9287971
					NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rotating band groove		0.40%	3.2	Gage
Minor					
201	Width of band groove minimum (min)		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.15	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 5	DRAWING NUMBER
4.4.2.2	Motor Body, Prior to Paint		9287971
			NEXT HIGHER ASSEMBLY 9287991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal Defects		100%	3.4.1	4.5.3/Visual
2	Magnetic Particle Inspection stamp missing		100%	3.2	Visual
Major					
101	Knurl missing, improper, inadequate		100%	3.2	Visual
102	Minor thread diameter, max.		100%	3.2	Gage
103	Thread pitch diameter		0.40%	3.2	Gage
104	Diameter of thread undercut		0.40%	3.2	Gage
105	Diameter of thread counterbore		0.40%	3.2	Gage
106	Length of small cavity adjacent to small end of exit cone		0.40%	3.2	Gage
107	Diameter of small cavity adjacent to small end of exit cone, max.		0.40%	3.2	Gage
108	Diameter of largest cavity		0.40%	3.2	Gage
109	Outside diameter of motor body, min. (2 places)		0.40%	3.2	Gage
110	Outside diameter of rear crests on rotating band (3 places)		0.40%	3.2	Gage
111	Outside diameter of second crest on rotating band		0.40%	3.2	Gage
112	Length of three rear band crests at basic distance (2 places)		0.40%	3.2	Gage
113	Length of two forward band crests at basic distance (2 places)		0.40%	3.2	Gage

NOTES: a. Examinations 137, 138, 139, 213 and 225 may be performed prior to knurling.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER 9287971
4.4.2.2	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Location of small basic diameter in exit cone		0.40%	3.2	Gage
115	Location of large basic diameter in exit cone		0.40%	3.2	Gage
116	Location of small basic outside diameter on boattail		0.40%	3.2	Gage
117	Location of large basic outside diameter on boattail		0.40%	3.2	Gage
118	True position of diameter of forward internal thread counterbore		0.40%	3.2	Gage
119	Runout of outside diameter of rotating band at three rear crests		0.40%	3.2	Gage
120	Runout of outside diameter of rotating band on second crest		0.40%	3.2	Gage
121	Runout of largest cavity with outside diameter		0.40%	3.2	Gage
122	Runout of small basic outside diameter on boattail with outside diameter of body		0.40%	3.2	Gage
123	Runout of large basic outside diameter on boattail with outside diameter of body		0.40%	3.2	Gage
124	Runout of diameter of small cavity adjacent to small end of exit cone with outside diameter of body		0.40%	3.2	Gage
125	Surface finish of thread counterbore improper		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 3 OF 5	DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET		9287971
			NEXT HIGHER ASSEMBLY 9287991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
126	Width of rotating band		0.40%	3.2	Gage
127	Angle of forward crest of rotating band		0.40%	3.2	Gage
128	Circular runout of forward crest of rotating band		0.40%	3.2	Gage
129	Diameter of rotating band at base of crest (4 places)		0.40%	3.2	Gage
130	Distance from shoulder at small end of exit cone to shoulder within cone		0.40%	3.2	Gage
131	Total distance from rear face of exit cone to flat face at bottom of large cavity		0.40%	3.2	Gage
132	Radius at bulkhead end of largest cavity improper (including visual for blending)		0.40%	3.2	Gage/Visual
133	Distance from rear face to rear of rotating band		0.40%	3.2	Gage
134	Ultrasonic inspection	(b)	100%	3.6	4.5.5
135	Ultrasonic inspection stamp missing		100%	3.6	Visual
136	Distance from end to thread undercut min.		0.40%	3.2	Gage
137	Runout of thread pitch diameter		0.40%	3.2	Gage
138	Length of thread counterbore		0.40%	3.2	Gage
139	Length of thread undercut, max.		0.40%	3.2	Gage
140	Mechanical properties (hardness)		100%		4.5.1.1
141	Mechanical properties (tension)	(c)			4.5.1.2
142	Chemical analysis	(d)			4.5.2
143	Magnetic particle test	(d)			4.5.3
144	Presence of surface voids, seams or cracks in rotating band		0.40%	3.4.2	Visual/Probe/Scale

NOTES:
(b) See paragraph 4.4.3.3.

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PARAGRAPH	TITLE	SHEET 4 OF 5	DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET		9287971
			NEXT HIGHER ASSEMBLY 9287991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
145	Depth of radial grooves in knurl, min. (Check three places, 120° apart)	(e)		3.13	4.5.9
146	Depth of undercut forward of rotating band, max.		0.40%	3.2	Gage
Minor					
201	True position of diameter of section between groove and shoulder within cone with outside diameter of body		0.65%	3.2	Gage
202	Inside diameter at rear of boattail		0.65%	3.2	Gage
203	Runout of rear inside diameter of boattail with outside diameter of body		0.65%	3.2	Gage
204	Inside diameter of section between groove and shoulder within exit cone		0.65%	3.2	Gage
205	Width of groove in large end of exit cone		0.65%	3.2	Gage
207	Location of groove in large end of exit cone (from rear face of exit cone)		0.65%	3.2	Gage
208	Slots missing, improper, insufficient or not equally spaced		0.65%	3.2	Visual
209	Surface finish of exit cone improper		0.65%	3.2	Visual
210	Radius at shoulder within large end of exit cone missing or improper		0.65%	3.2	Visual
211	Taper on inside of boattail and radii at rear of bulkhead between exit cone and boattail missing, improper or improperly balanced		0.65%	3.2	Visual

NOTES: (c) See paragraph 4.4.3.1.2
(d) See paragraph 4.4.3.2
(e) See paragraph 4.4.3.13

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET				9287971
					NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
212	Radius in small cavity adjacent to small end of exit cone missing or improper		0.65%	3.2	Visual
213	Chamfer on thread counterbore missing or improper		0.65%	3.2	Visual
214	Diameter of band seat adjacent to rotating band		0.65%	3.2	Gage
215	Distance from rear face of band to basic dimension on each crest of band		0.65%	3.2	Gage
216	Thickness of base (bulkhead)		0.65%	3.2	Gage
217	Profile of rotating band including any damage to any crest		0.65%	3.2	Visual
218	Runout of outside diameter to the rear of the rotating band to outside diameter forward of the band		0.65%	3.2	Gage
219	Outside diameter of exit cone section		0.65%	3.2	Gage
220	Runout of outside diameter of exit cone section		0.65%	3.2	Gage
221	Length of band seat adjacent to rotating band		0.65%	3.2	Gage
222	Length of largest cavity with thread		0.65%	3.2	Gage
223	Total length of motor body		0.65%	3.2	Gage
224	Minor thread diameter, min.		100%	3.2	Gage
225	Evidence of poor workmanship		0.65%	3.15	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Motor Body After Paint				9287971
					NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of Motor Body (max)		100%	3.2	Gage
102	Paint on rotating band, knurl or on any threads		100%	3.2	Visual
103	Diameter of cavity adjacent to small end of exit cone, min.		0.40%	3.2	Gage
104	Any imperfection of forward crest of band		0.40%	3.2	Visual
105	Salt spray test	(a)		3.2	4.5.7
Minor					
201	Protective finish with total exposed or damaged area in excess of 1/4 square inch on interior surfaces		0.65%	3.2	Visual
202	Coating with total exposed or damaged area in excess of 1/4 square inch on interior surfaces		0.65%	3.2	Visual
203	Marking missing, misleading, incomplete or illegible		0.65%	3.2	Visual
204	Protective coating thickness, min.	(b)		3.2	4.5.6
205	Evidence of poor workmanship, including rust or corrosion		0.65%	3.15	Visual

NOTES:
(a) See Paragraph 4.4.3.9.
(b) See Paragraph 4.4.3.10.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Inert, Nozzle		9287963
			NEXT HIGHER ASSEMBLY 9287991

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Runout of end surfaces (2 places)		0.40%	3.2	Gage
103	Largest inside diameter diverging section (2 places)		0.40%	3.2	Gage
104	Runout of tapered I.D. with outside diameter (2 places)		0.40%	3.2	Gage
105	Chamfer missing or obviously improper		0.40%	3.2	Visual
106	Surface finish of inside profile		0.40%	3.2	Visual
107	Throat diameter		0.40%	3.2	Gage
108	Compression test load	5		3.8.1	4.5.8.1
109	Voids in nozzle insert		100%	3.8.2	4.5.8.2
110	Orientation of grain flow		100%	3.2	4.4.3.6.2
Minor					
201	Runout of throat diameter with outside diameter		0.65%	3.2	Gage
202	Total length		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.15	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287964
4.4.2.5	Support, Grain				NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter	(a)		3.2	Gage
102	Inside diameter	(a)		3.2	Gage
103	True position of inside diameter	(a)		3.2	Measure Distance from O.D. to I.D. (4 places)
104	Overall thickness	(a)		3.2	Gage
105	Flatness of large surface	(a)		3.2	Gage
106	Presence of voids	(b)		3.9	Visual
Minor					
201	Flat on outside diameter missing or improper	(a)		3.2	Visual
202	Radius improper	(a)		3.2	Gage
203	Evidence of poor workmanship	(a)		3.15	Visual

NOTES: (a) See Paragraph 4.4.3.4.1/4.4.3.4.2.
(b) See Paragraph 4.4.3.4.3.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287986
4.4.2.6	Cushion, Aft				NEXT HIGHER ASSEMBLY 9287991
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter	(a)		3.2	Gage
102	True position of inside diameter	(a)		3.2	Verify Die
103	Inside diameter	(a)		3.2	Gage
104	Thickness	(a)		3.2	Gage
Minor					
201	Workmanship	(a)		3.15	Visual

NOTES: (a) See 4.4.3.5.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287991
4.4.2.7	Rocket, Motor Bonding Assembly, 8 Inch: M650				NEXT HIGHER ASSEMBLY 9288025
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving ground test	(a)		3.7	4.5.5
Major					
101	Nozzle insert missing or not securely bonded (prior to assembly of grain support)		100%	3.2	Manual
102	Nozzle insert not flush or below (prior to assembly of grain support)		0.40%	3.2	Gage
103	Grain support missing, inverted or not secure		0.40%	3.2	Visual/Manual
104	Aft cushion missing, improper or not secure		0.40%	3.2	Visual
105	Excess adhesive in motor cavity		0.40%	3.2	Visual
106	Adhesive in delay seat after curing of grain support and cushion		0.40%	3.2	Visual
107	Improper application of adhesive to grain support or motor body prior to assembly		0.40%	3.2	Visual
108	True position of nozzle insert, after adhesive cure		0.40%	3.2	Visual
109	Evidence of phosphate coating in grit blasted area, prior to assembly of grain support (applies if grit blast is prior to phosphate)		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.15	Visual

NOTES: (a) See Paragraph 4.4.3.8.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9288025
4.4.2.8	Motor Body Assy.				NEXT HIGHER ASSEMBLY 9287992
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Cap-Plug missing on rocket nozzle		0.65%	3.2	Visual
202	Protective plug missing		0.65%	3.2	Visual
203	Inhibitor missing (prior to assembly of cap-plug)		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.15	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48323C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Packaging				8837835
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Pallet damaged (unserviceable)		0.65%	3.2	Visual
202	Improperly packed		0.65%	3.2	Visual
203	Strapping missing, loose or broken		0.65%	3.2	Visual
204	Marking missing, incorrect or illegible		0.65%	3.2	Visual
205	Box split or broken		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Mechanical properties of motor body.

4.4.3.1.1 Hardness. In order to determine uniformity of heat treatment, each body after application of the welded overlay and heat treatment shall be subjected to a hardness test with a Brinnell Hardness Tester or any other method approved by the procuring activity. Hardness impressions shall be made on the body located as shown on Dwg. 9287971. The 3 hardness readings thus obtained shall be averaged and used as the basis for selecting the tension test samples of paragraph 4.4.3.1.2.

4.4.3.1.2 Tension test. Two banded motor bodies from the same heat treated inspection lot shall be selected for this test (one body from the highest hardness reading and one motor body from the lowest hardness reading). From each of these motor bodies three specimens shall be tested to determine compliance with yield strength and elongation requirements as follows:

a. One from the sidewall directly under the rotating band in a longitudinal direction.
b. One from the sidewall as close to the open end as possible in a longitudinal direction.
c. One from the bulkhead (heavy wall at the closed end which is perpendicular to the axis of the part). This test bar shall be taken from the center of the wall such that its axis is on a radial line perpendicular to the longitudinal center line of the part.

If a specimen fails to meet the requirements of the drawing, the heat treat lot shall be rejected subject to retest as specified in 4.4.3.1.3.

4.4.3.1.3 Retest. If the high hardness motor body fails the yield strength requirement, the entire heat treatment inspection lot shall be rejected. In the event the lowest hardness motor body fails to meet the yield strength requirement, higher hardness values may be selected and tested in accordance with 4.4.3.1.2 in order to establish the acceptable minimum hardness. Once this minimum hardness has been established, all motor bodies with hardness values below the new minimum shall be rejected. If the low hardness motor body fails the elongation requirement, the entire heat treatment inspection lot shall be rejected. In the event the highest hardness motor body fails to meet the elongation requirement, lower hardness values may be selected and tested in accordance with 4.4.3.1.2 in order to establish the acceptable maximum hardness. Once this maximum hardness has been established, all motor bodies with hardness values above the new maximum shall be rejected.

4.4.3.2 Copper welded overlay chemical analysis, seat metal defects and cratering. One motor body per eight hour shift, or fraction thereof, per welding machine shall be subjected to the

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tests of paragraph 4.4.3.2.1 thru 4.4.3.2.5. The sample shall be serialized to maintain identity with the motor bodies represented. Rejection shall be as specified in paragraph 4.4.3.2.6.

4.4.3.2.1 Chemical analysis. Copper welded overlay chips machined from the motor bodies selected in accordance with paragraph 4.4.3.2 shall be used for this test. (See paragraph 4.5.2).

4.4.3.2.2 Metal defects - copper welded overlay seat. The motor bodies selected in accordance with paragraph 4.4.3.2 shall be used for this test (see paragraph 4.5.3).

4.4.3.2.3 Metal defects - copper welded overlay seat (subsurface). The motor bodies selected in accordance with paragraph 4.4.3.2 shall be examined for indications at the 7.810 inch and 7.770 inch diameter (see paragraph 4.5.3).

4.4.3.2.4 Base metal cratering - copper welded overlay seat. The motor bodies selected in accordance with paragraph 4.4.3.2 shall be used for this test.

4.4.3.2.5 Base metal cratering - copper welded overlay seat (subsurface). The motor bodies selected in accordance with paragraph 4.4.3.2 shall be examined at the 7.840 inch and 7.70 inch diameter.

4.4.3.2.6 Rejection. If one or more motor bodies fail any test of paragraph 4.4.3.2.1 thru 4.4.3.2.5, fail to comply with the requirements of paragraph 3.4.2 thru 3.4.4 or 3.5.1 or 3.5.2.1 the motor bodies in the shift/machine represented by the test sample shall be rejected.

4.4.3.3 Ultrasonic inspection.

4.4.3.3.1 Motor body inside diameter prior to ultrasonic inspection. The inside diameter and surface finish shall be measured (I.D. gaged and finish-visually) in the area of the rotating band for conformance to requirement, specified in 3.6.4 according to the following inspection plan:

Inspect the first 100 motor bodies manufactured on a 100 percent basis. After meeting this, inspection may be reduced to a sampling plan in accordance with MIL-STD-105, Level I, and AQL of 0.40%.

4.4.3.3.2 Ultrasonic inspection. This inspection shall be performed 100 percent after copper deposition and heat treating, with the inside diameter machined as specified in 3.6.4. Failure of the motor body to meet the requirements of 3.6 shall be cause for rejection of that motor body. This inspection may be conducted prior to finish machining.

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4.4.3.4 Support, Grain.

4.4.3.4.1 Initial sampling. The first pieces from each mold shall be inspected for all defects listed in 4.4.2.5 until ten consecutive parts from each mold have been found acceptable.

4.4.3.4.2 Subsequent sampling. One piece per each 1000 produced from each mold will be inspected in accordance with 4.4.2.5. If any major defect is found the production represented by the sample will be rejected. Production will be stopped and the mold corrected. Sampling for this mold will return to 4.4.3.4.1 when production begins again. If any minor defect is found corrective action will be taken.

4.4.3.4.3 Void examination. The first piece from each mold shall be cut in half and examined for the requirement in 3.9. If the piece is defective each piece that follows shall be cut in half and examined until an acceptable part is produced. Additionally, one piece from each 200 produced from each mold shall also be cut in half and examined in accordance with 3.9. If any piece is defective that quantity represented by the sample shall be rejected.

4.4.3.5 Cushion, Aft.

4.4.3.5.1 Initial sampling. The first pieces from each die shall be inspected for all defects listed in 4.4.2.6 until ten consecutive parts from each die have been found acceptable.

4.4.3.5.2 Subsequent sampling. One piece per each 100 produced from each die will be inspected in accordance with 4.4.2.6. If any major defect is found the production represented by the sample will be rejected. Production will be stopped and the die corrected. Sampling for this die will return to 4.4.3.5.1 when production begins again. If any minor defect is found corrective action will be taken.

4.4.3.5.3 True position of inside diameter. Each die shall be inspected for the true position of inside diameter at the start and conclusion of a shift. If the die fails to comply with the specified true position all parts made from that die shall be rejected.

4.4.3.6 Nozzle insert.

4.4.3.6.1 Compression tests. A sample of five (5) nozzle inserts from each lot shall be compression tested in accordance with 4.5.8. If any piece fails to meet the minimum compression load requirement of 3.8.1 the lot shall be rejected.

4.4.3.6.2 Grain flow. A certificate of compliance shall be submitted with each lot that the grain flow of the nozzle insert is in accordance with the drawing. The contractor shall show evidence

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that adequate controls are being maintained in processing the nozzle insert from the billet or sheet to finished component to assure proper grain flow direction.

4.4.3.6.3 Radiographic inspection. This test shall be conducted 100 percent. The test shall be conducted in accordance with 4.5.8.2. If any nozzle insert fails to meet the requirements of 3.8.2, that piece shall be rejected.

4.4.3.7 Tool control dimensions.

a. Dimensions marked tool control shall be gaged at the beginning or end of each working day (24 hrs.), during First Article Inspection and for the first component after each tooling change, adjustment or repair. Where destruction of the component is necessary to gage such dimensions, measurements of the tool may be substituted providing the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production.

b. When the contractor has satisfied (a) above, visual inspection of tool control dimension(s) will be permitted. Visual inspection shall be performed every two (2) hours for each machine in operation. If any tool control dimension(s) is in doubt during visual inspection, the dimension(s) shall be immediately gaged to determine acceptability. Corrective action will immediately be instituted by the contractor whenever a defective tool control dimension is found.

4.4.3.8 Proving ground tests.

4.4.3.8.1 First article test. Twenty (20) motor body assemblies from the first article sample shall be subjected to this test. If any of the sample assemblies fail to meet the requirements of 3.7, the first article sample shall be rejected. Range and deflection shall be recorded for information only. The test shall be conducted as specified in 4.5.5.

4.4.3.8.2 Production test. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, twenty motor body assemblies shall be selected at random from each lot for this test. If any of the sample assemblies fail to meet the requirements of 3.7, the lot shall be rejected. After three consecutive lots have complied with the acceptance criteria, ten samples shall be selected at random from each lot and tested. If any sample motor body assembly fails to meet the requirements of 3.7, the lot shall be rejected. Range and deflection shall be recorded for information only. The test shall be conducted as specified in 4.5.5.

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4.4.3.9 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.10 Protective coating - thickness. Two samples shall be selected at random to represent each four hour period, or less, of continuous paint production. The samples shall be inspected using the equipment in accordance with 4.5.6. The sample motor bodies shall be inspected for paint coating thickness at a minimum of four separate locations on each motor body. Any motor body failing to meet the applicable requirements shall be classed defective and every motor body painted since the last acceptable sample shall be rejected.

4.4.3.11 Hydrogen embrittlement relief. The contractor shall comply with the requirements of TT-C-490. For verification the contractor shall submit a certificate of compliance that motor bodies have been relieved of hydrogen embrittlement.

4.4.3.12 Stress relief (if required). The contractor shall comply with the requirements of drawing 9287871. For verification, the contractor shall submit a certificate of compliance that the motor bodies have been stress relieved.

4.4.3.13 Depth of knurl.

4.4.3.13.1 Initial inspection. The first pieces from each tool, forming the radial groove, shall be inspected for depth until one part has been accepted (see 6.10).

4.4.3.13.2 Subsequent sampling. If the life of the tool forming the radial grooves is 50 parts or less, the last part formed by each tool shall be inspected. If tool life is greater than 50 parts, an additional inspection of one part from each 50 shall be performed. If the knurl depth is less than .012 inches, production represented by that sample shall be rejected, production shall be stopped and corrective action taken. Sampling for any tool producing a rejected part shall revert to 4.4.3.13.1 when the tool is returned to production (see 6.10).

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6. herein.

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4.5 Methods of inspection.

4.5.1 Mechanical properties of motor body.

4.5.1.1 Hardness. The hardness test shall be conducted in accordance with ASTM, Method E 103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding to a sufficient depth, .015 inches minimum, to remove scale and get below the decarburization zone. The test may be conducted on the finish machined surface.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically displayed.

d. The Hardness Tester shall use a standard 10 MM Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 300Kg. The amount of penetration of the indentor (10MM ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

e. Three hardness readings shall be taken 120 degrees apart.

4.5.1.2 Tension. The tension test shall be conducted in accordance with ASTM, Method E8 with the largest sub-standard size round possible.

4.5.2 Chemical analysis. The iron content of the copper welded overlay shall be determined by chemical analysis of chips machined from the overlay. The sample shall be taken by removing chips from a diameter of 8.195 - .005 to 8.145 - .005 and dissolving all the chips. Chemical analysis shall be conducted in accordance with FED-STD-151 except the sample shall not be cleaned with a magnet.

4.5.3 Magnetic particle test. The inspection shall be in accordance with specification MIL-I-6868 using the continuous, wet, fluorescent method. Both circular and longitudinal magnetizations are required. During inspection of the motor body and copper welded overlay the suspension shall first be applied to all surfaces of the entire motor body. During inspection for indications at the 7.810

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inch diameter and 7.770 inch diameter and for maximum base metal cratering at the 7.840 inch diameter only the machined area beneath the rotating band must be covered with suspension. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 milliliters (ml) of solids per 100 mls suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. After draining the excess suspension the motor body shall be inspected under ultraviolet light for the presence of defects. The adequacy of the inspection process shall be checked at the start of daily production and at eight hour intervals thereafter using a standard in accordance with Dwg. 9280261. All plugs in the standard shall be clearly defined for a full 360 degrees both inside and out. If the inspection process is determined to be inadequate the condition shall be corrected and all motor bodies accepted since the last standard check shall be reinspected. The requirement of specification MIL-I-6868 to check system effectiveness with a steel ring specimen shall be waived. Components which exhibit indications which are suspected of being shallow surface defects may have material removed to a maximum of .005 inches subject to the following restrictions:

a. These provisions of the removal of additional material shall not apply to the inspections at the 7.810 inch, 7.770 inch and 7.840 inch diameter. In addition these provisions shall not apply to indications on thread surfaces.

b. Involved areas shall be blended with adjacent surfaces to insure that no sharp edges, steps or gouges remain.

c. The involved areas shall be independently checked for verification that all drawing dimensions (wall thickness, variations, etc.) are within acceptable limits.

d. The components shall be again subjected to magnetic particle inspection for verification of defect removal.

4.5.3.1 Operator qualification. For operation of magnetic particle equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of magnetic particle equipment, personnel shall be qualfied and certified in accordance with MIL-STD-410, Level II.

4.5.4 Ultrasonic Inspection. Each banded motor body shall be inspected utilizing equipment and procedures in accordance with Envelope Dwg. 9280353.

4.5.4.1 Automated Acceptance. If automated c-scan interpretation is to be utilized, a standard to verify correct interpretation of the requirements in 3.6.3 shall be designed by the contractor. This standard shall be used with the calibration standard (see Envelope Dwg. 9280353) for all verification checks.

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4.5.4.2 Verification and operational procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including description of equipment and standards (Contractor Design) and operation procedures (see 6.3) No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by running a verification check on the calibration standard(s) at the start and finish of each production shift, as well as at four (4) hour intervals during production. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. A log book shall be maintained by the operator on each ultrasonic system which, as a minimum, documents the processing of standards, number of items inspected, changing of transducers, instrumentations, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

After completion and acceptance of the ultrasonic tests and the final verification test has passed, clearing the previously inspected motor bodies, each acceptable motor body shall be marked in accordance with Dwg. 9287971. Any rejected motor bodies shall be clearly marked as such and removed from the acceptable motor bodies.

4.5.4.3 Operator qualification. For operation of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of ultrasonic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II. Periodic verification checks required may be conducted by the Level I operator, however, any equipment adjustment required as a result of a verification check shall be made by Level II qualified personnel.

4.5.5 Proving ground. These tests shall be performed at a government proving ground in accordance with the applicable Acceptance Test Procedure. The sample motor body assemblies selected in accordance with 4.4.3.8 shall be inert loaded and assembled and fired at ambient temperature in a M201 Series cannon with muzzle brake using a propelling charge adjusted which produces an average pressure of 47,000 + 1000, -1500 psi chamber pressure based on copper pressure gages. The propelling charges shall be uniformly temperature conditioned prior to the test. Pressure-time traces and differential pressure shall be obtained for information only for each round fired by means of a spindle tap in the breech of the cannon and a piezo-gage at the projectile base location. Adequate photographic and instrumental coverage shall be provided to assure compliance with 3.7. Range and deflections shall be recorded when possible for informational purposes only. Recovery and examination of metal parts for any abnormal condition shall be accomplished for information only.

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4.5.6 Protective coating - thickness. The protective coating thickness test shall be accomplished using commercially available equipment.

4.5.7 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectiles utilizing the equipment and procedures specified by FED-STD-141 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.4.3.9 and test samples shall be dried in accordance with Drawing 9287971.

4.5.8 Nozzle insert

4.5.8.1 Compression test. The sample nozzle inserts shall be placed in a suitable testing machine capable of maintaining a 0.02 inch per minute head speed. The nozzle insert shall be placed in the machine with the chamfered end down and the load applied axially. If the nozzle insert meets the minimum load requirement, the load shall be applied until the insert cracks or breaks up and the load recorded for informational purposes.

4.5.8.2 Radiographic test. The nozzle inserts shall be radiographed with the cavity down and the radiographic beam parallel to the longitudinal axis. Radiographic inspection shall be in accordance with MIL-STD-453. The penetrameter shall be designed so as to assure a 2 percent sensitivity and the film density shall be 1.50 minimum.

4.5.8.2.1 Operator qualification. For operation of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.9 Depth of knurl. The depth of the radial groove in the knurl shall be checked using a dial indicating depth gage or approved equivalent.

5. PACKAGING

5.1 Packaging requirements. Minimum Military Package: Pack one Rocket Motor Body Assembly in a RSC Style, "B" Flute, Type CF, domestic class, SW variety, Grade 200 min. Spec PPP-B-636, Fiberboard box. Inside dimensions of fiber box to be 8-7/16 x 8-7/16 x 10-7/8 inch high (Tolerance on dimension + 1/16 inch). Rocket Motor Body assembly shall be placed in the fiberboard box with the protective plug (knurled end) down. 8.0 x 8.0 x 1/16 inch thick fillers shall be added on sides and top for tight packing.

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Flaps on fiber box to be sealed in accordance with Spec PPP-B-636. The packed fiber box shall be placed in a Type 1, Class E, Spec MIL-B-117 barrier bag. Prior to sealing bag, exhaust all excess air. Size of bag to be 18-1/2 inch. min open face x 23-1/4 inch min length. Seal to be located at edge of bag to provide for a minimum of 2 subsequent seals. The seals shall comply with the heat seal test specified in Spec MIL-P-116.

5.2 Packing. Minimum Military Pack - 20 packaged motor bodies shall be placed vertical in a 5 x 4 pattern on a wood pallet in accordance with Dwg. 8837835. Pallet to be packed and strapped in accordance with Dwg. 8837835.

5.3 Marking.

5.3.1 Fiber box. Marking of fiber box and bag to be with black stencil ink Spec TT-I-1795. Label may be used on fiber box. Size of letters 1/8 inch min. Marking on the fiber box to be centrally located on one side. Marking on bag to be so positioned that it appears on the side of the box. Mark as follows:

1 - Rocket Motor Body Assembly
Lot No._____

5.3.2 Pallet. Marking of pallet to be in accordance with Dwg. 8861520. Descriptive Nomenclature: 20 - Rocket Motor Body Assembly Lot No._____.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 8-Inch, M650, RA Projectile.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection procedures and equipment designs for approval See MIL-A-48078. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Center, ATTN: AMSMC-QAR-I, Dover, New Jersey 07801-5001.

6.4 It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

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6.6 Proving ground test summary.

Test	Sample Size (see 4.4.3.11)	Requirements
Security	*20 10	No break-up or metal parts separation (see 3.5)

*Applicable only for First Article Test and first three lots.

6.7 Submission of ballistic test data. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data shall be forwarded to Commander, ARDC, Dover, New Jersey 07801-5001, ATTN: AMSMC-QAR-Q and SMCAR-LCU-SA.

6.8 Start zone. The starting weld zone as referenced to herein is defined as a zone 1/2 inch on each side of the starting pass centerline.

6.9 Copper welded overlay seat. The rotating band seat width specified on drawing 9287971.

6.10 Knurl inspection. Due to the nature of the "knurl", some minor wear of the crests may result from normal handling subsequent to machining. Inspection for knurl effective depth should be conducted as a process check immediately after machining.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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